

Topic: Packages — Self-Learning

Q. Discuss the concept of Packages in database programming (such as PL/SQL). Explain the difference between a Package Specification and a Package Body, and list their advantages.

📌 Definition to Remember

A **Package** (in Oracle PL/SQL) is an advanced database programming construct that **logically groups related Stored Procedures, Functions, Variables, Cursors, and Exceptions** into a single named module. It is analogous to a "Class" or "Namespace" in object-oriented programming, consisting of a **Package Specification** (public interface) and a **Package Body** (implementation).

1. Structure of a Package

A package is strictly divided into **two separate components**:

PACKAGE SPECIFICATION
(Declares: procedures, functions,
variables, constants, cursors)

← PUBLIC Interface
(What the package can do)

PACKAGE BODY
(Contains actual executable code
+ private variables/procedures)

← PRIVATE Implementation
(How it does it)

A. Package Specification (Public Interface)

- Declares what is available to the outside world — the **public API**.
- Defines signatures of procedures/functions WITHOUT the actual code.

```
CREATE OR REPLACE PACKAGE HR_Manager AS
    PROCEDURE Hire_Employee(emp_id INT, name VARCHAR);
    FUNCTION Get_Total_Employees RETURN INT;
END HR_Manager;
```

B. Package Body (Implementation)

- Contains the actual **executable code** for the procedures/functions declared in the spec.
- Can also contain **private** variables and procedures NOT declared in the specification.

```

CREATE OR REPLACE PACKAGE BODY HR_Manager AS
  PROCEDURE Hire_Employee(emp_id INT, name VARCHAR) IS
  BEGIN
    INSERT INTO Employee VALUES (emp_id, name);
  END;
  FUNCTION Get_Total_Employees RETURN INT IS
  BEGIN
    -- implementation
  END;
END HR_Manager;

```

2. Advantages of Packages

Advantage	Explanation
Modularity	Groups related routines (e.g., all HR operations in one <code>HR_Package</code>)
Encapsulation	Body hides private logic; only spec is visible to users
Performance	Entire package loaded into memory on first call; subsequent calls = no disk I/O
Overloading	Same procedure/function name can have multiple signatures with different parameters

★ Must-Write Points (for 10 marks)

1. A Package groups related procedures, functions, variables, and cursors into one module.
2. It has two parts: **Package Specification** (public interface) and **Package Body** (implementation).
3. Specification declares the procedure/function signatures — the public API.
4. Body contains the actual code + private variables not visible in the specification.
5. **Performance**: entire package loads into memory on first call — faster subsequent access.
6. **Encapsulation**: users see only the specification; body logic is hidden (information hiding).
7. **Overloading**: same name can have multiple versions with different parameter lists.

⚡ Quick Recall

Package → Groups related DB objects → Specification (Public API) + Body (Private Implementation) → Advantages: Modularity + Encapsulation + Memory Performance + Overloading